DOCUMENT 4:
DEMA CNC CASE STUDY

This document provides an overview of the DEMA CNC Case Study. It serves as a complementary document to the "Document 3: DTAF AI PIPELINE- CNC CASE STUDY" document. This document is meant to be used in conjunction with both the DTAF AI Pipeline Tool and the standalone DEMA software. DEMA is one tool that can be used as part of the Data Feasibility Layer of DTAF, with the alternative to DEMA being the Data Feasibility Checklist. This guide is intended for parties that have chosen to use DEMA in place of the Data Feasibility Checklist and have established DEMA methodology and software license agreements with Auburn University [US Patent 12,131,134]. The DEMA Software is free for all U.S. federally funded work and available for licensing for private sector use. For more information on access to the DEMA methodology, please reach out to Dr. Allison Ledford at anb0051@auburn.edu.

This guide is organized into four sections: Section 1 presents an overview of the DEMA methodology and defines its key terminology. Section 2 describes how to transition from the DTC Capability Periodic Table of the AI Pipeline into Option 2 (DEMA) of the Data Feasibility Layer. Section 3 explains how to use the DEMA software to create a complete DEMA dataset. Section 4 then explains how to input the DEMA dataset into the AI Pipeline to move into the final stage of DTAF to output the DT Build Guide. Figure 1 highlights in red the section of the DTAF AI Pipeline Workflow that this document covers.

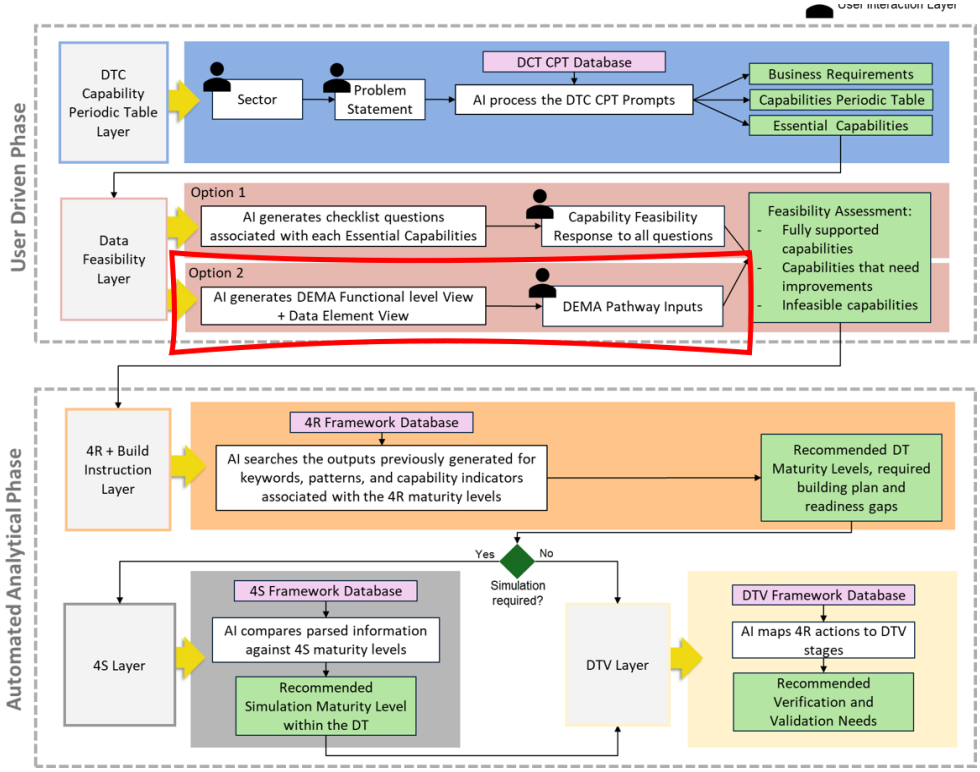


Figure 1: DEMA Portion of the DTAF AI Pipeline Workflow.

Section 1: DEMA Methodology Overview and Terminology

Data Element Mapping and Analysis (DEMA) provides a systematic approach for identifying, capturing, and mapping data flows in complex systems. DEMA helps organizations understand where their data truly originates and where it ultimately ends up. Unlike traditional methods that focus on documents or high-level processes, DEMA dives deeper by examining data at the data element level, the smallest meaningful units, such as part numbers, approval dates, or design dimensions. By doing this, DEMA uncovers hidden or manual data exchanges that often disrupt system efficiency and reliability. Upon completion, the DEMA output (Data Element Level View) can be translated into a graph structure. From this graph structure, improved graph structures representing the proposed DT can be created to calculate before-and-after metrics of efficiency. Therefore, a complete DEMA data set can be used to holistically capture the current state of the candidate DT system, assess how "ready" the current state is for DT implementation using the 4R approach, and calculate the return on investment (ROI) in terms of reduced labor hours to show the estimated cost savings of any proposed Digital Twin configuration.

The DEMA process follows three steps. The first step, the Functional-Level Mapping, identifies the major functional areas and sub-activities that make up an organization's workflow. This step establishes the foundation for understanding where data is created, used, and exchanged. The Functional Level maps show how departments interact and how processes align with organizational goals, helping the analyst trace every piece of data to its functional purpose. The second step, Data Vessel Level Mapping, dives deeper by documenting the movement of containers, called data vessels, across these functions. Data vessels include spreadsheets, reports, CAD files, emails, or even handwritten notes. It can be anything that stores or transfers data. In this step, DEMA identifies how these vessels are shared, where they are stored, and who has access to them. By capturing these flows, organizations can see how data travels between people and systems. The third step, Data Element Level Mapping, is where each data vessel identified in the second step is broken down into its smallest components, individual data elements, and relationally linked across instances. These elements might include measurements, serial numbers, inspection results, or timestamps. Mapping these elements allows the analyst to trace exactly how data is created, modified, and reused across systems and departments. By following these three steps, DEMA transforms hidden, fragmented data into a transparent, traceable network that supports digital engineering ROI calculation for Digital Twins (DTs).

Section 2: DTC Capability CPT → Data Feasibility Option 2: DEMA

The output of the first step in the DTAF AI Pipeline Workflow is the Digital Twin Consortium Capabilities Periodic Table (DTC CPT) for the user's proposed system. Each of the capabilities identified in the CPT can be used to create a DEMA Functional Map. The user can do this manually, or with the assistance of generative AI of their choosing. To do this using generative AI assistance, please reference the "Data Feasibility Layer - Option 2: Guided DEMA Process for 4R Feasibility Assessment" document (specifically Section 4.2). Based on the Gen AI output, the user can edit the Functional Map according to their needs and preferences. The key is that the user verifies that the Functional Map satisfies the capabilities in the CPT and that each functional activity is independent of one another (not overlapping). For a complete example of using

Generative AI to generate a DEMA Functional Map based on the DTC CPT output, please see Section 4.3 of the “Data Feasibility Layer - Option 2: Guided DEMA Process for 4R Feasibility Assessment" document.

Section 3: Creating a DEMA Data Set Using the DEMA Software

The first step in creating a DEMA Data Set is to fill in the information in the DEMA software for the Functional Level View (DEMA Step 1). To accomplish this, the user selects either a new or existing project (Figure 2), and then selects "Functional Level” from the software project home page (Figure 3), and then follows the prompts to input the Functional Activities created using the instructions from Section 1 of this report. Figures 4 - 7 demonstrate this process for the CNC Machining Process case study. After completing the Functional Level View, the user navigates back to the project home page and selects Data Vessel Level (see Figure 8). The user then follows the prompts to fill in the information for the Data Vessel Level View (Figures 9 - 11). The comprehensive Data Vessel Map can be viewed by selecting the map icon at any point during the data vessel mapping process (see Figure 12).

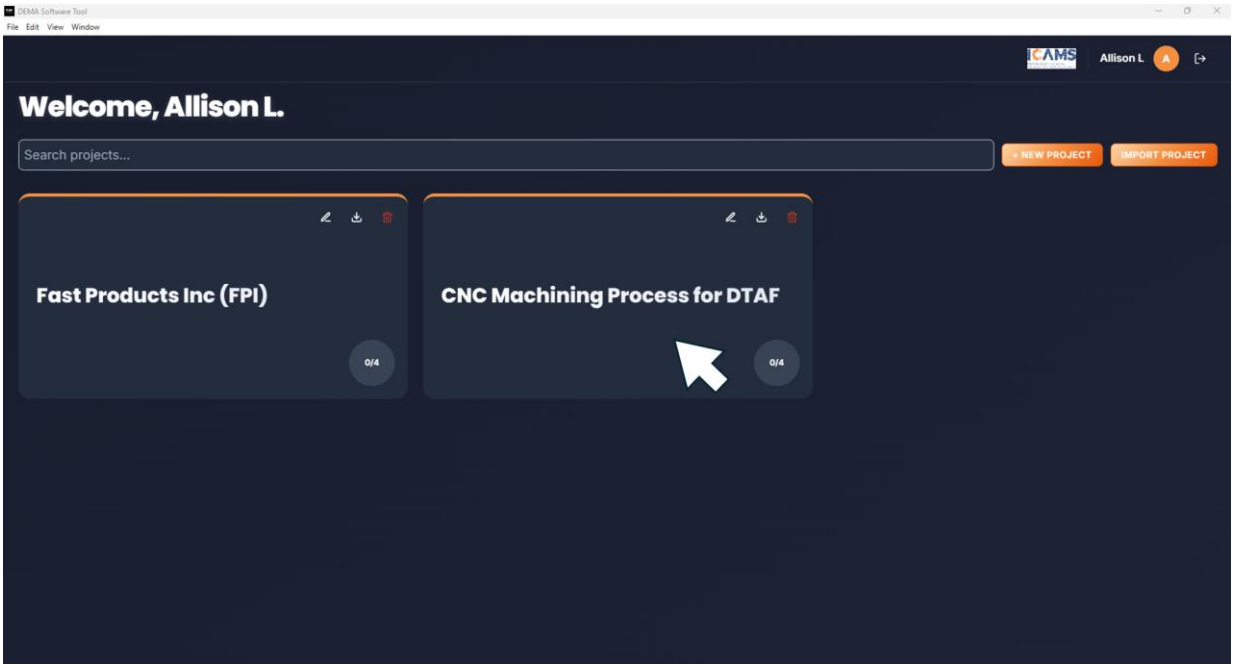


Figure 2: Choose a New or Existing Project in the DEMA Software.

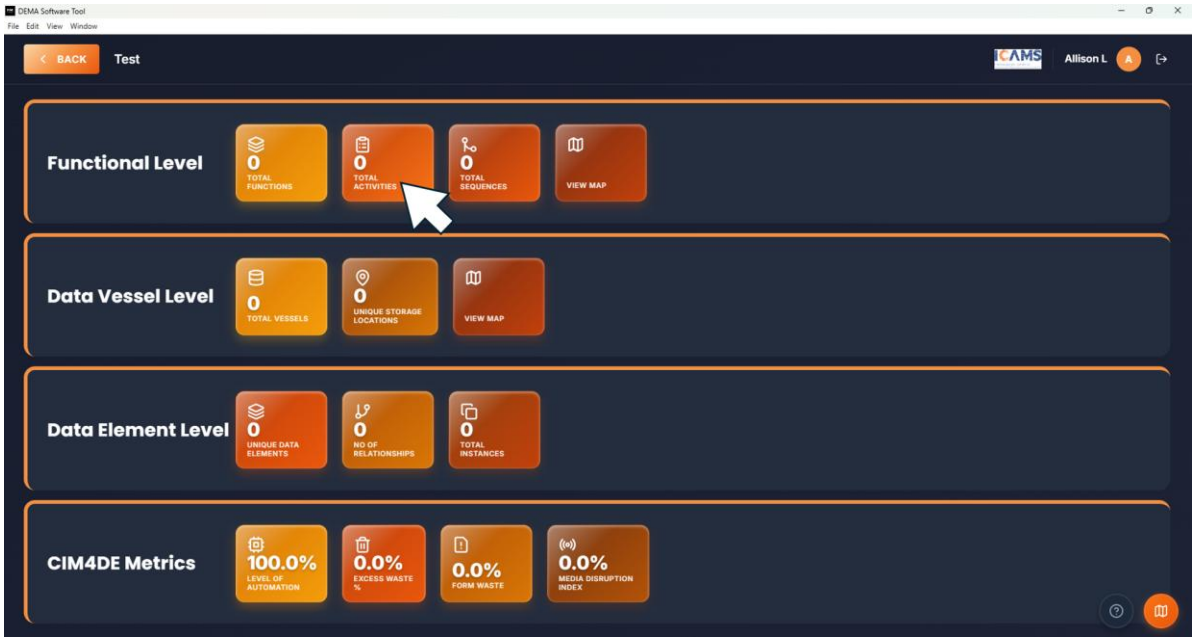


Figure 3: Select the Functional Level View.

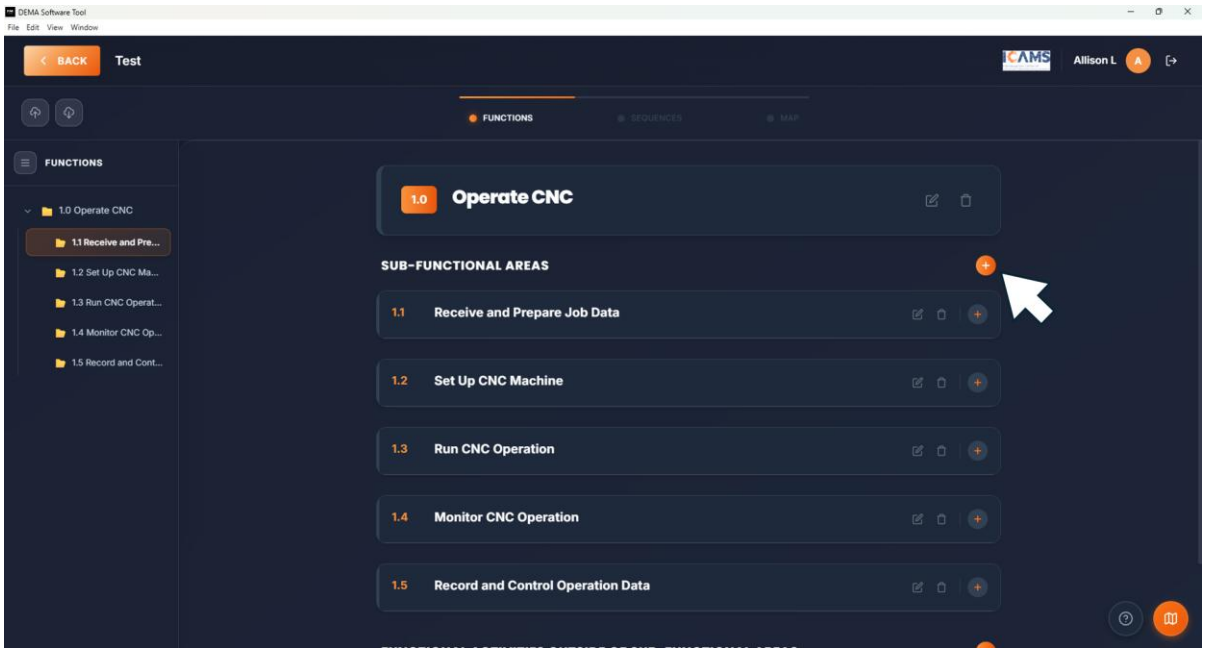


Figure 4: Add the Functional Areas and Sub-Functional Areas.

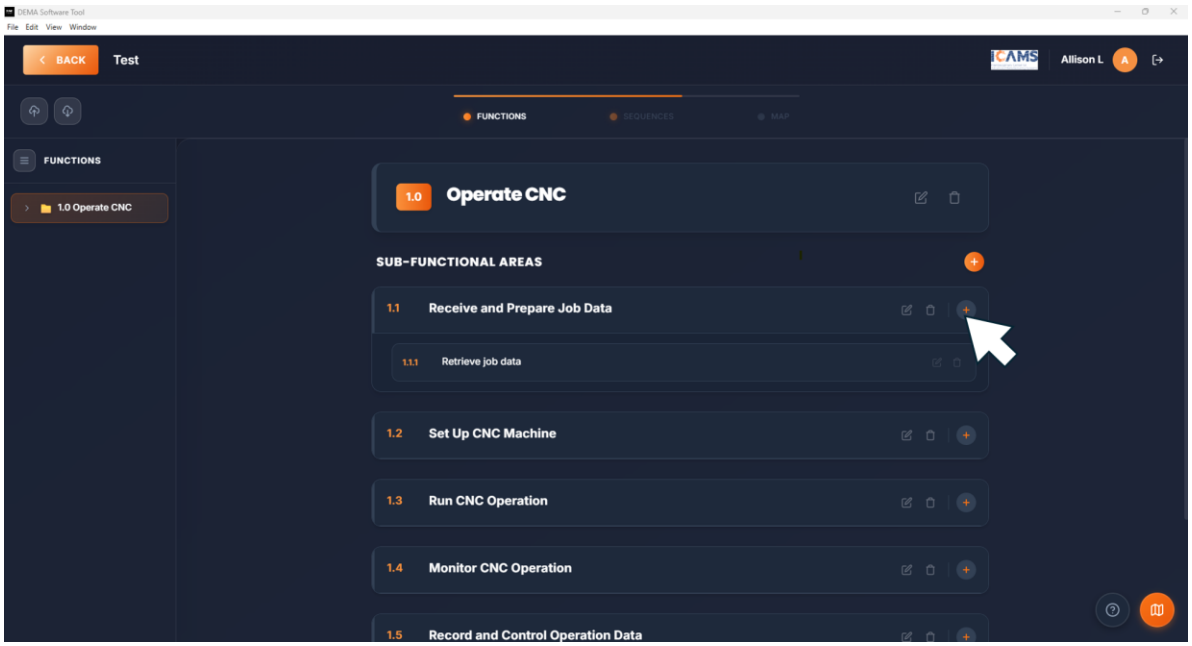


Figure 5: Add the Functional Activities.

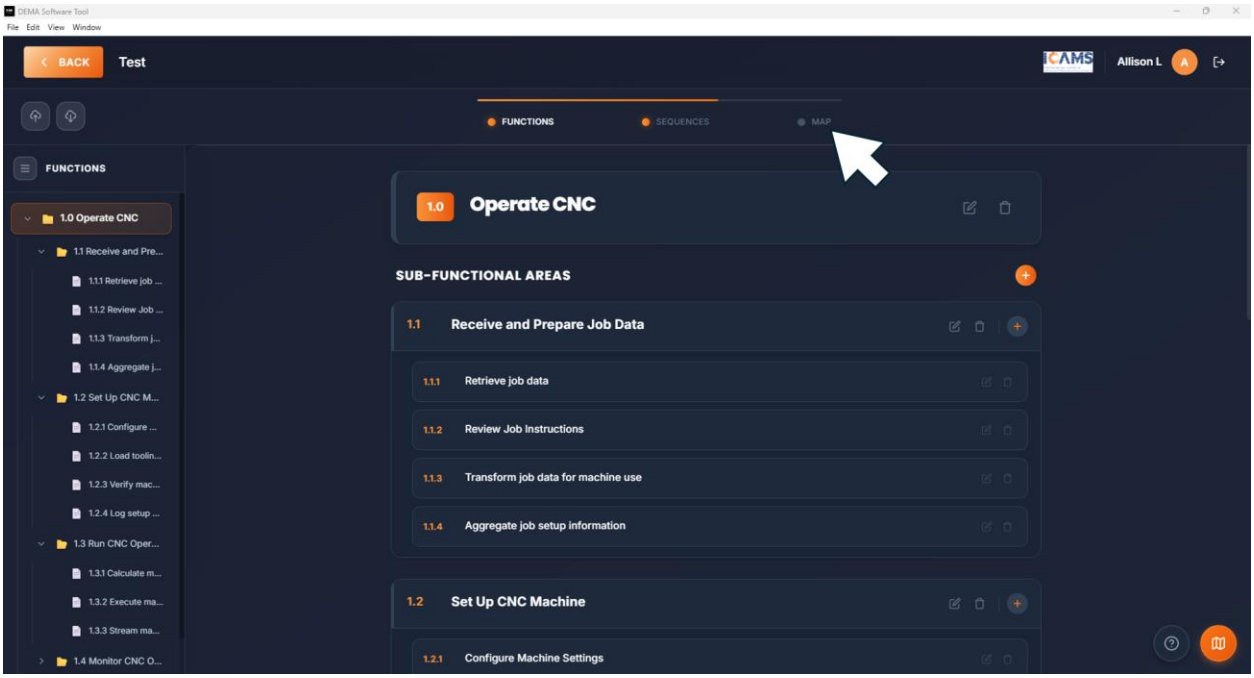


Figure 6: Complete the Functional Level View.

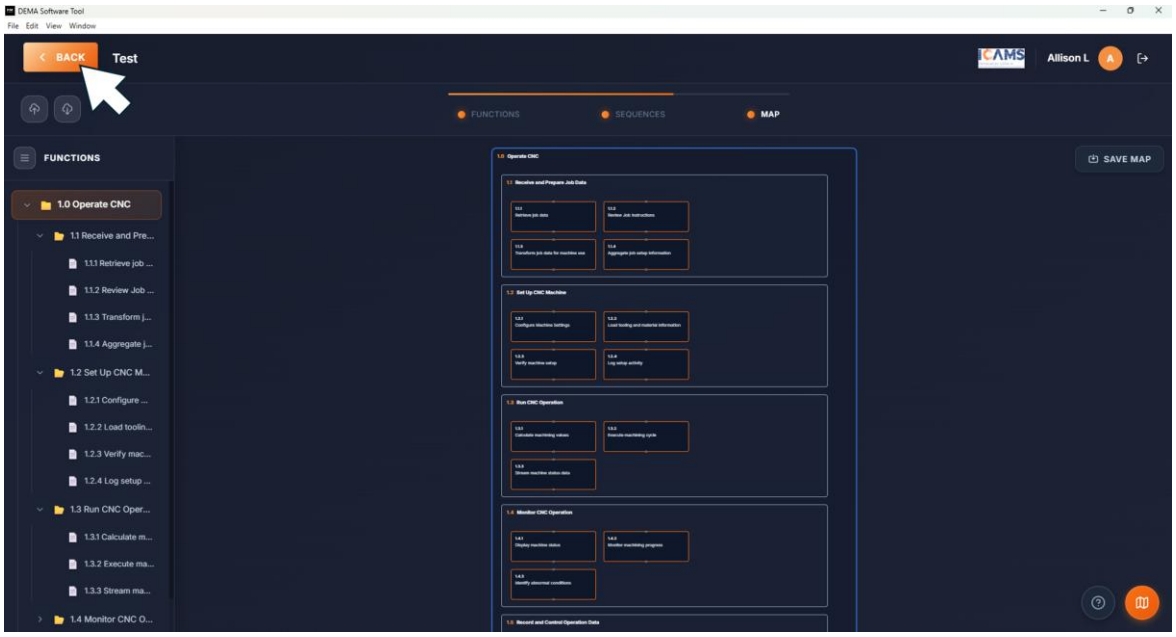


Figure 7: The Complete Functional Level View Map for the CNC Machining Case Study.

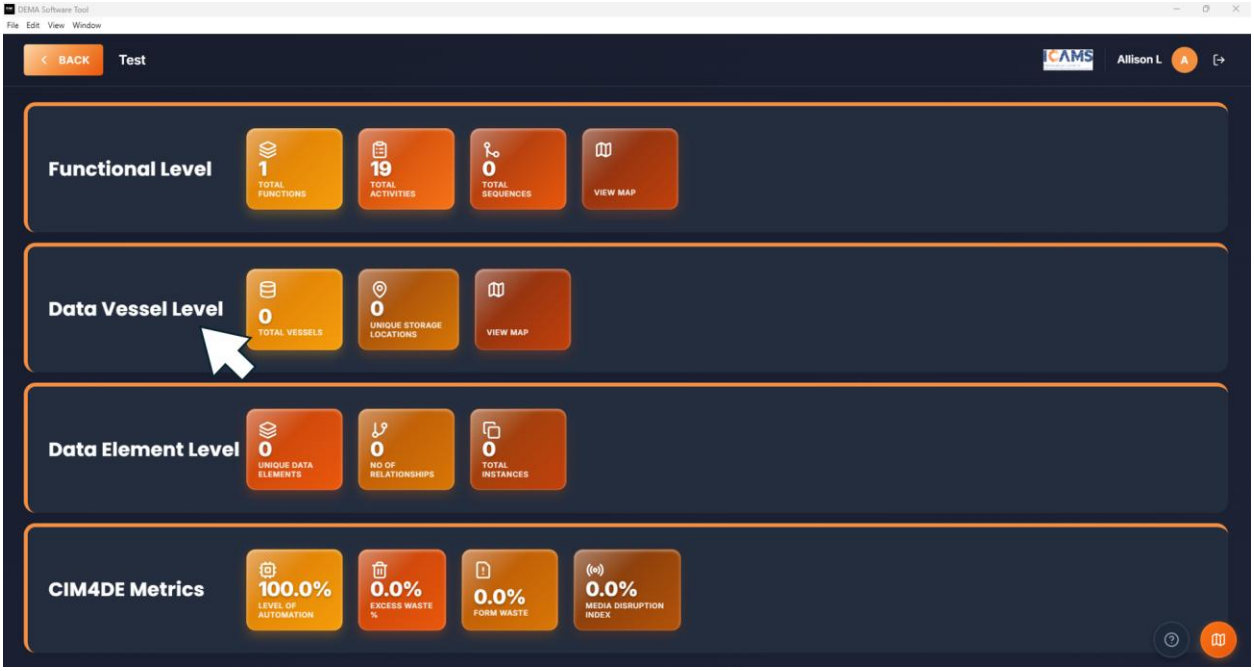


Figure 8: Select the Data Vessel Level View.

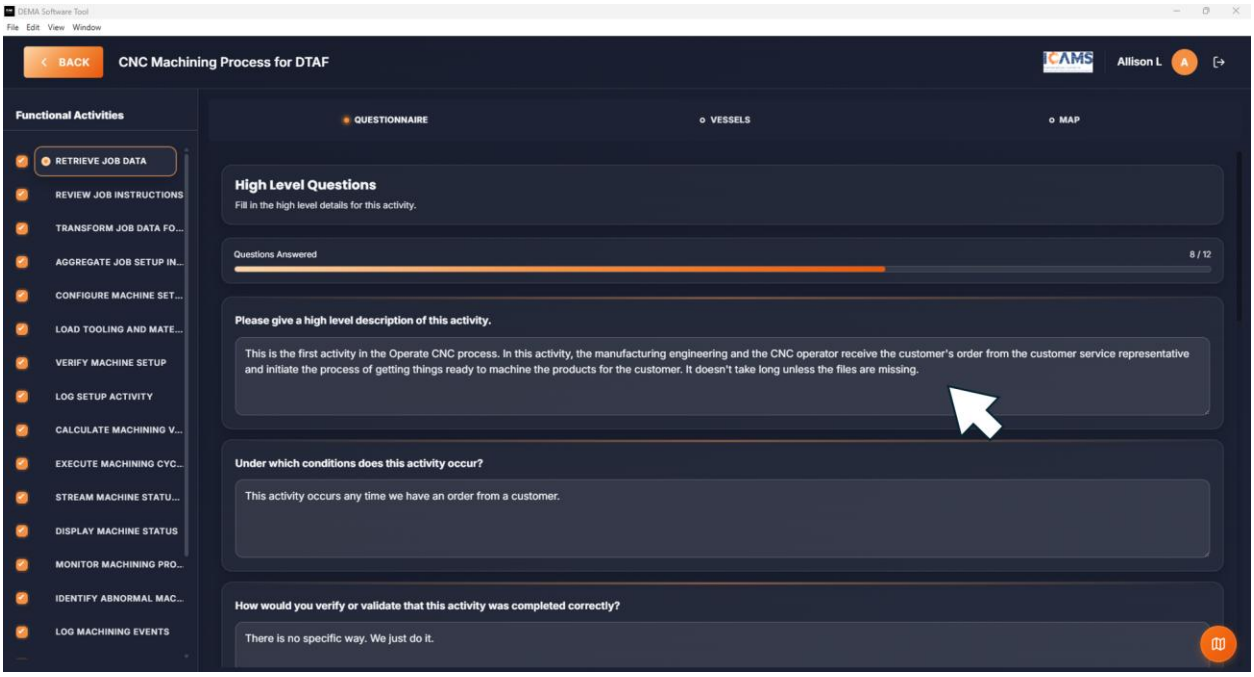


Figure 9: Answer the High-Level Questions for Each Functional Activity.

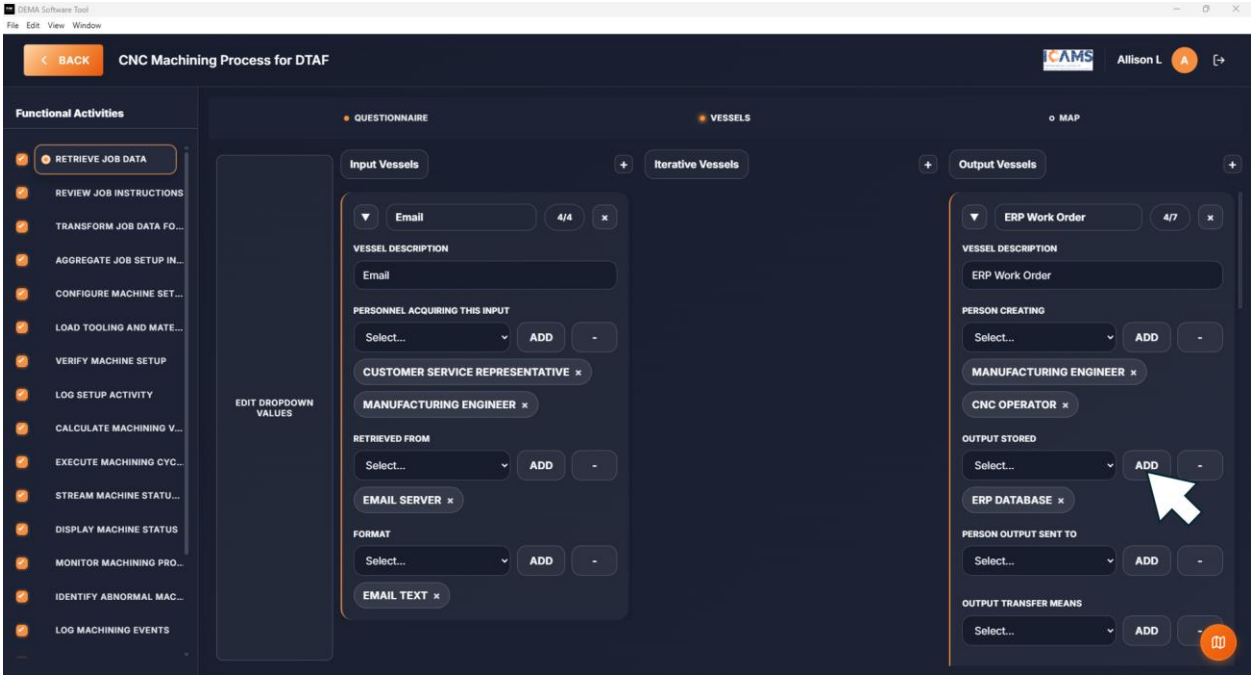


Figure 10: Answer the Data Vessel Level Questions for Each Activity.

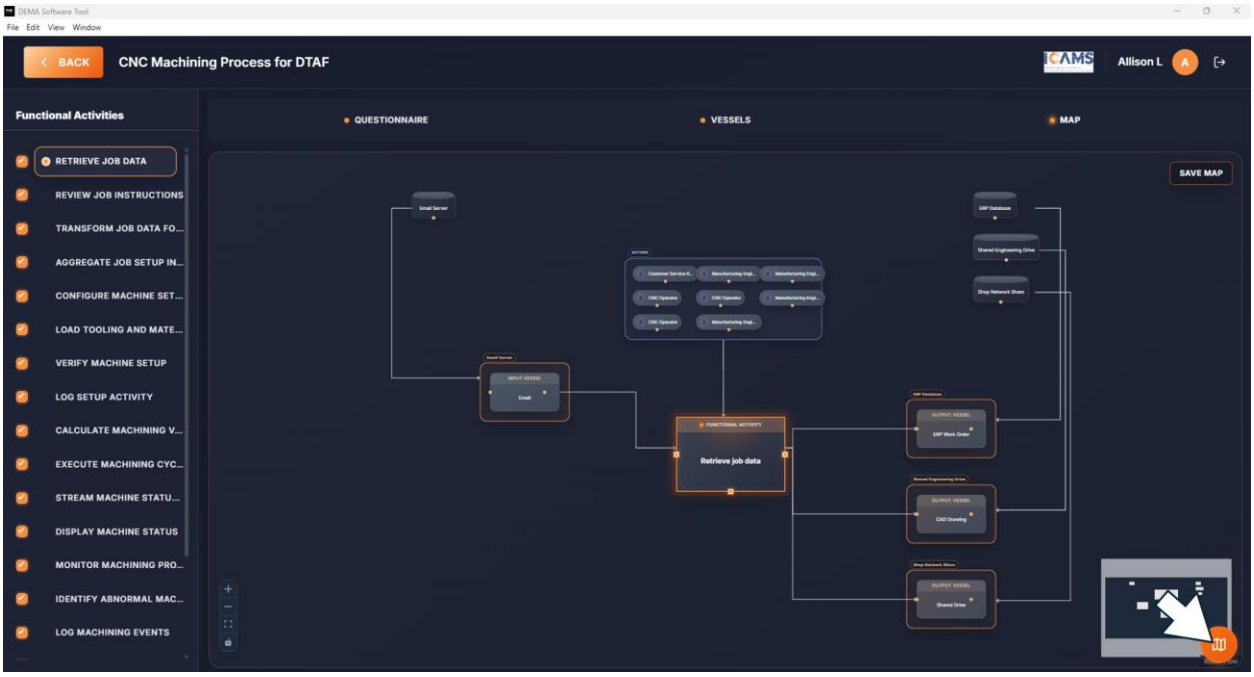


Figure 11: View the Data Vessel Map for Each Activity.



Figure 12: View the Comprehensive Data Vessel Map to View Progress.

After completing one or more activities in the Data Vessel Level View, the user navigates back to the project home page and selects the Data Element Level (see Figure 13). The user can then follow the prompts to fill in the information for the Data Element Level View. To assign data elements to each data vessel, the user clicks on the name of the functional activity they want to start with and then clicks on any vessel in the Data Vessel Map that appears (see Figure 14), and then assigns new or existing data elements to that vessel (Figure 15). Once all data elements are added for the functional activity, the user selects the “Define Relationships" button (see Figure 16). The page that allows the user to create data element instances and links between instances will appear (see Figure 17). Once the DEMA Data Element Level View is complete, it is ready for download. To download the Data Element Level View at any point in the process, navigate to the Data Element Table page (see Figure 18) and select the “Download” button (see Figure 19).

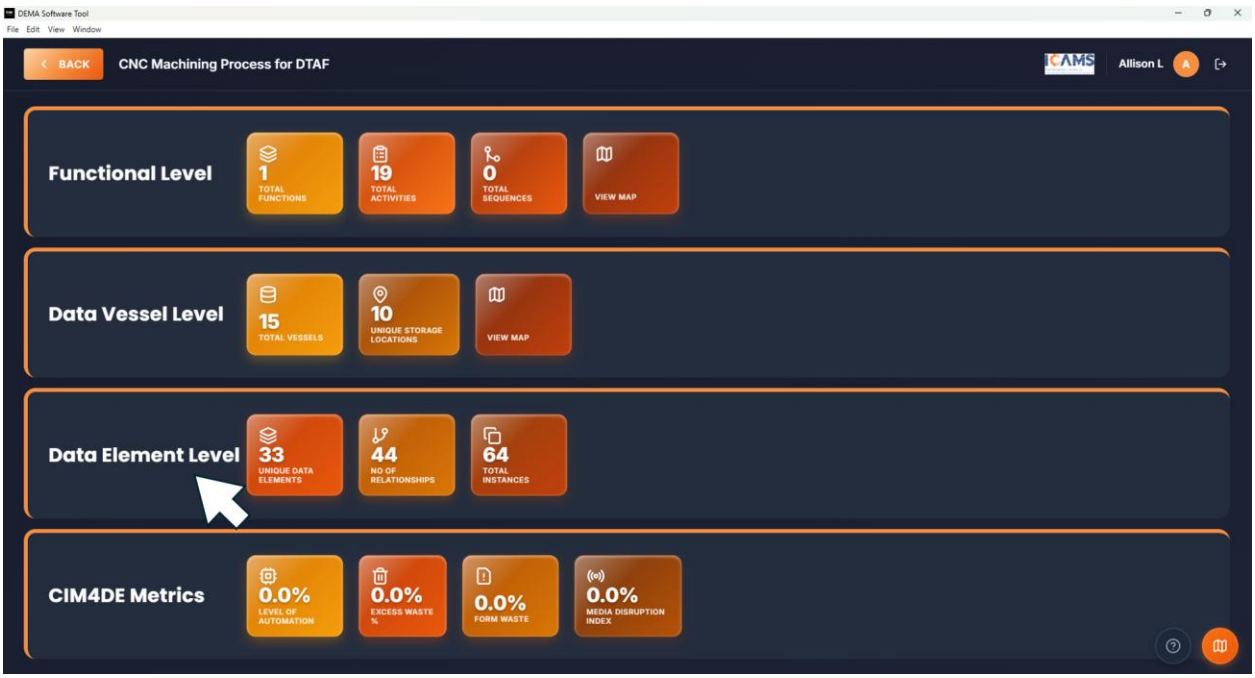


Figure 13: Select the Data Element Level View.

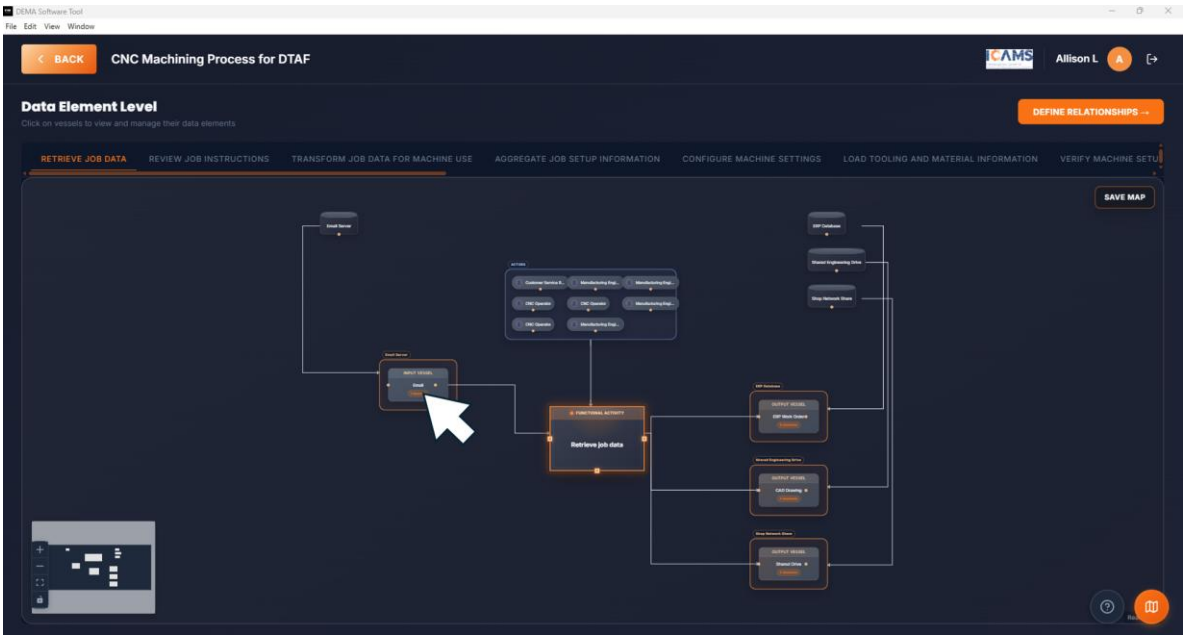


Figure 14: Selecting the Data Element Level View.

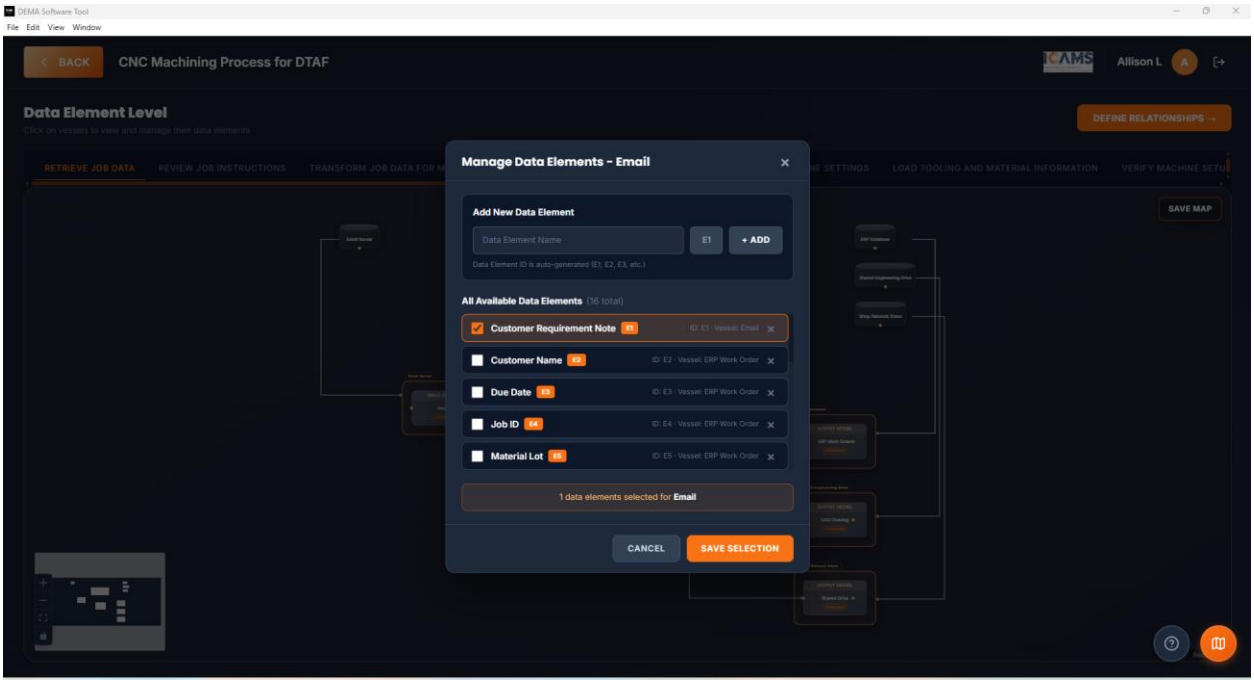


Figure 15: Manage Data Element Assigned to Each Vessel.

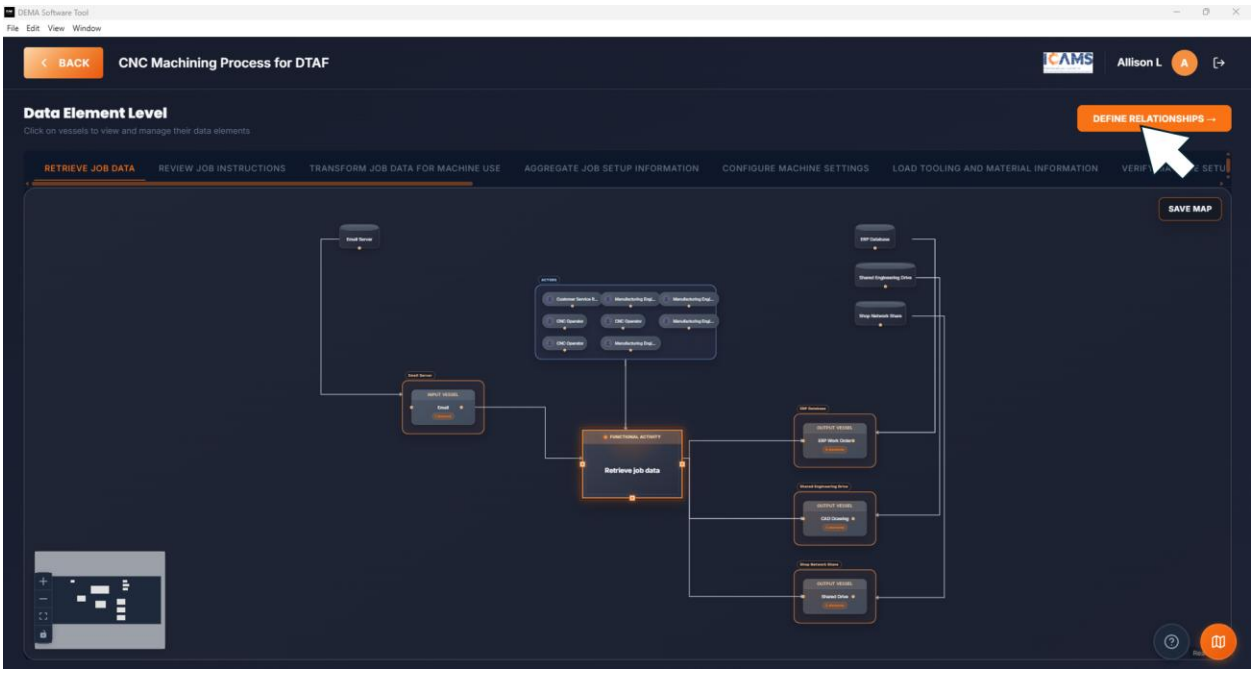


Figure 16: Navigate to “Define Relationships”.

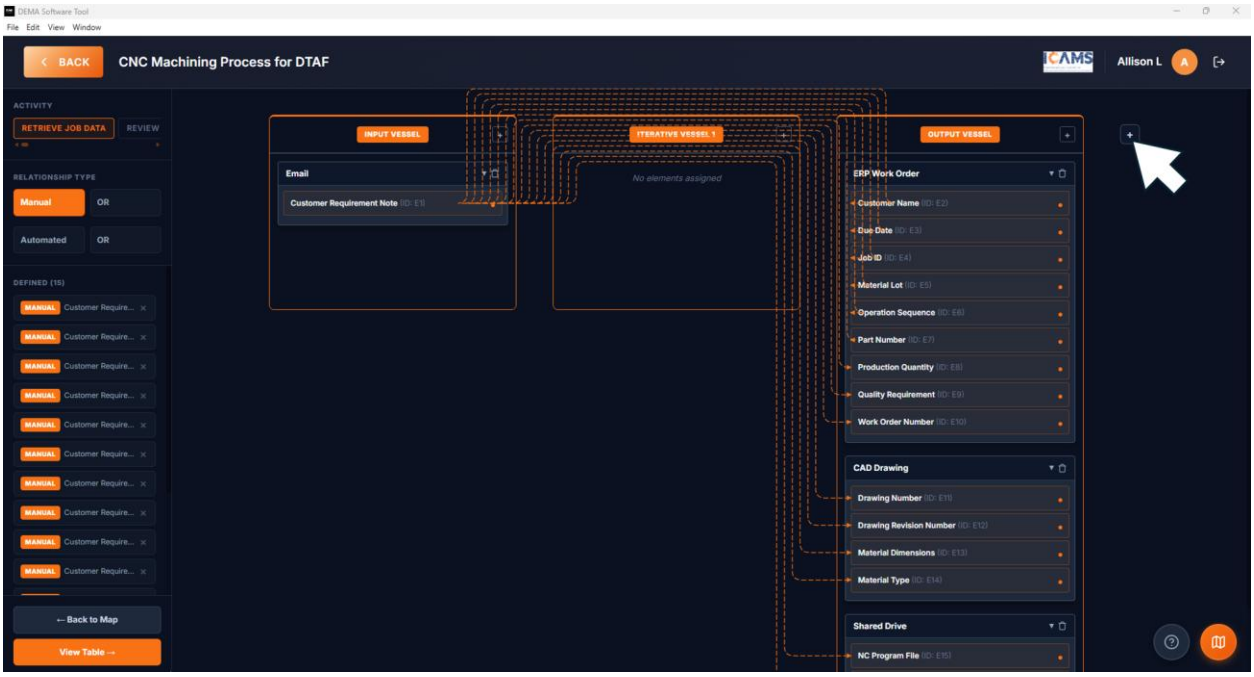


Figure 17: Define Data Element Instances and Relationships.

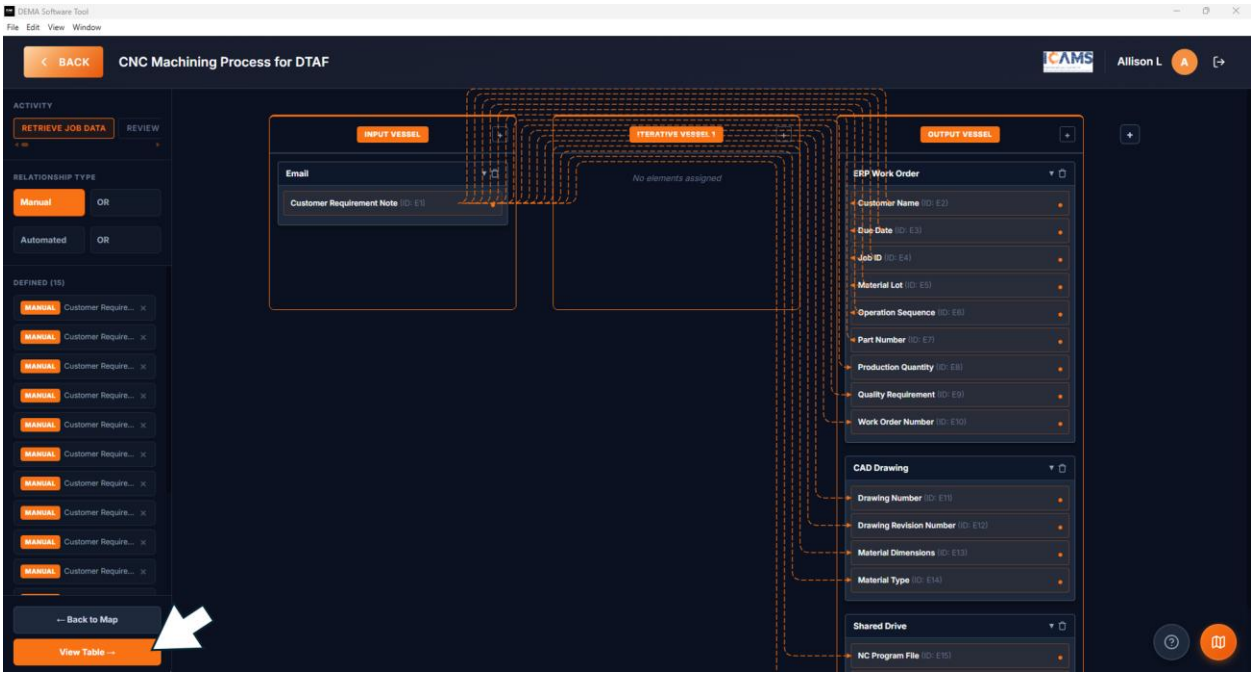


Figure 18: Navigate to Data Element Level View Table.

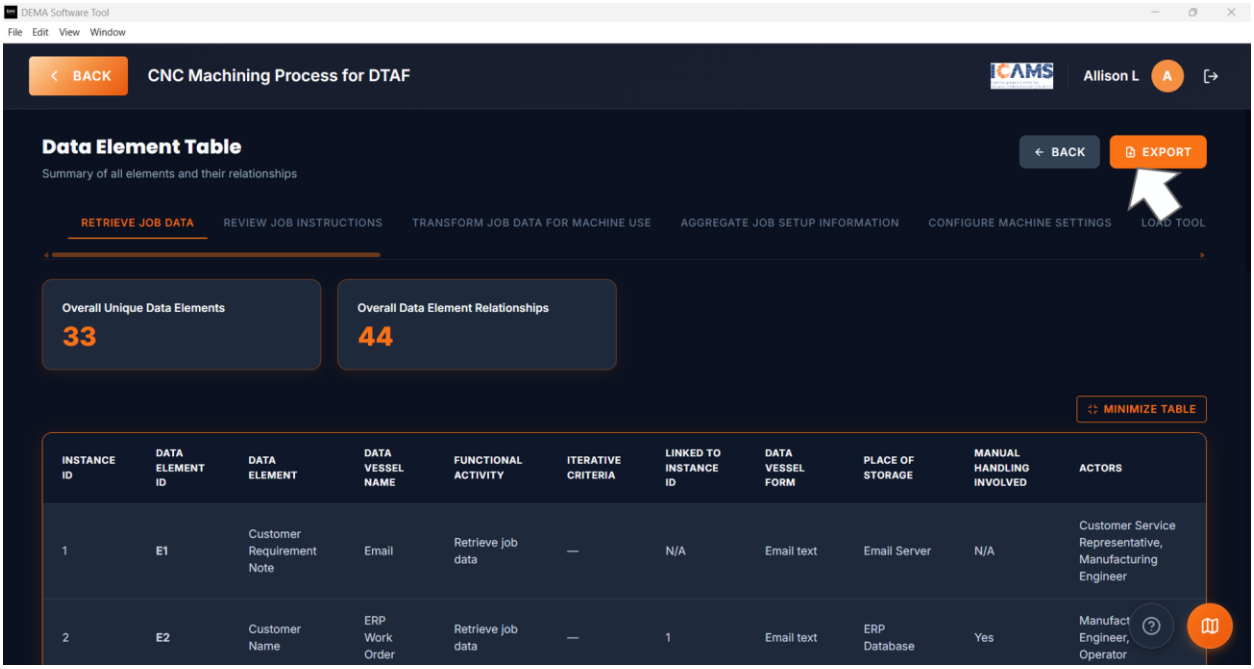


Figure 19: View Data Element Level View Table.

To view the associated metrics of efficiency of the systems as captured in the Data Element Level View, the user can navigate to the project home page and select “CIM4DE Metrics” (see Figure 20). These metrics can be used to assess the efficiencies of the current state of the system’s data flows (see Figure 21). Once a new system architecture for the DT has been defined, the user can

go through the DEMA software again to update the data element relationships based on how the new system will be set up and compare the metrics of the current state (before DT implementation) to the proposed new state (the DT system). This way, the user can ensure that the proposed DT architecture has enough of an ROI to justify implementation.



Figure 20: Select CIM4DE Metrics.

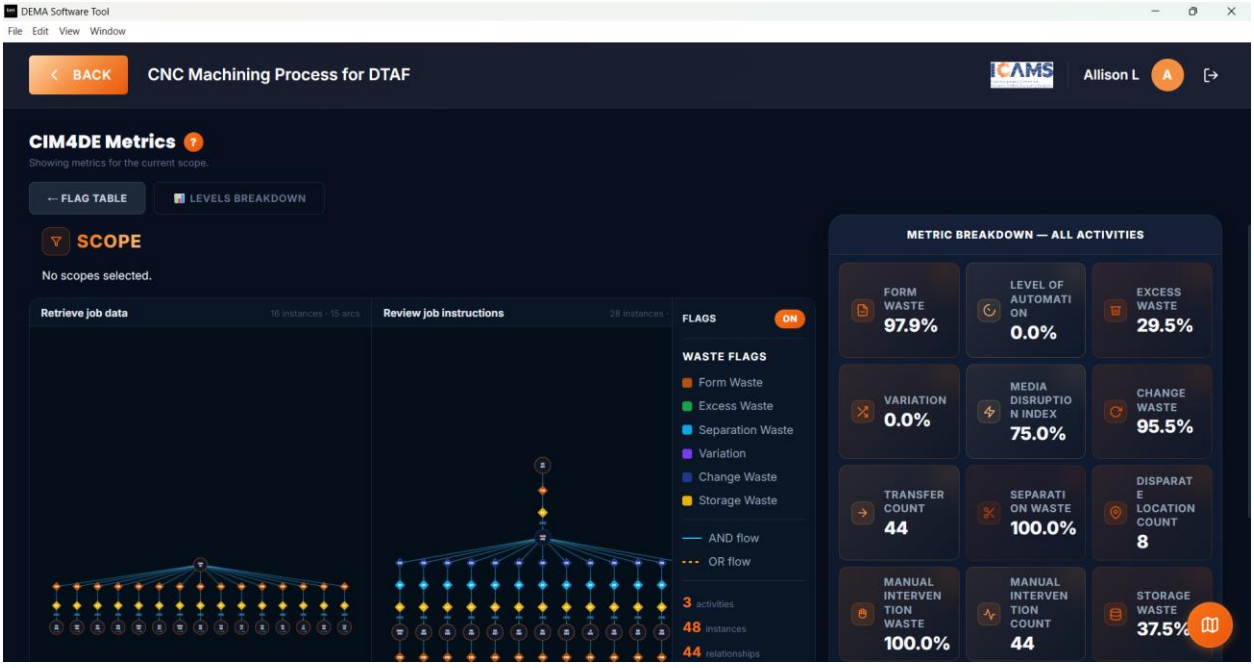


Figure 21: Example CIM4DE Metric Calculations

Section 4: Inputting the DEMA Data Set into the DTAF AI Pipeline

The ability to input a Data Element Level View data set from the DEMA software and upload it directly into the DTAF AI Pipeline is under development. Updates will be added to the DTAF AI Pipeline GitHub as these capabilities become available. Upon completion, this functionality will allow the use of the DTAF AI Pipeline to produce a DT Build Guide based upon the DEMA results instead of the Data Feasibility Checklist. Future work may include estimated ROI calculations incorporated into the DT Build Guide based on DEMA inputs.